

A. Configuration-Route

DUT issues configuration write



FPGA emits ordered control event



Host updates authoritative image and affected BAR route

happens-before



Shadow update and matching ACK



DUT write completes

B. Submission-DMA

DUT writes queue entry then rings doorbell



FPGA accepts BAR event



Adapter observes stable entry and translates DMA references

happens-before



Translated command becomes visible



Physical doorbell is forwarded

C. Data-Completion

Physical backend completes DMA



Payload and writebacks become visible in DUT memory

happens-before



Adapter publishes driver-visible completion state



Pending state is updated



Virtual INTx level may change